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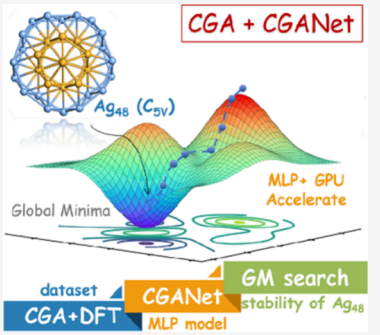
Accelerating Global Search of Large-Sized Silver Clusters Using Cluster Graph Attention Network

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ABSTRACT: Great efforts have been devoted to understanding the stability and reactivity of silver clusters, which usually depend on geometric structures, electronic configuration, and cluster size. Despite the fact that the jellium model and Wulff construction rule have successfully rationalized the stable clusters with "magic number" behavior, some experiments imply that silver clusters with 48 valence electrons also possess puzzling enhanced stability. In this work, using a recently developed deep learning technology, i.e., cluster graph attention network (CGANet), combined with a homemade comprehensive genetic algorithm (CGA) program, we searched the global minimum (GM) structures of Ag_n (n = 30−60) clusters with graphics processing unit acceleration, whose efficiency is about 2 orders of magnitude higher than that of the conventional density functional theory (DFT) calculations. GM structures and some representative isomers are reported at each size, revealing the competitive structural patterns based on truncated octahedra and icosahedra as well as the icosahedra-based layer-by-layer growth mode of large-sized Ag clusters. Most importantly, the size-dependent evolution behavior of structural and electronic properties of Ag_n (n = 30−60) clusters can successfully explain the observed stability at Ag_{48}. Therefore, CGANet provides a powerful tool for rapidly exploring the potential energy surface of atoms with an accuracy comparable to that of DFT.

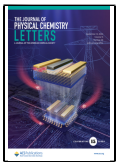


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Clusters, as an intermediate state bridging atoms and condensed matter, have received considerable research attention in catalytic, biological, microelectronic, optical, and magnetic applications.[1−3] From a technological point of view, the unique advantage of clusters lies in the control of the number and composition of atoms, geometric structure, and electronic properties with atomic-level precision. Clusters of coinage metals (i.e., Cu, Ag, and Au) are particularly interesting due to their relatively simple electronic configuration ($d^{10}s^1$) as well as their fascinating photochemical and catalytic properties.[4−9] Among them, Ag clusters exhibit a rather complicated size-dependent evolution of geometries, as signified by the strong competition of different structural patterns.[10−12] It was found that small neutral silver clusters Ag_n (n = 3−6) usually adopt planar structures,[13] while a structural transition from planar to three-dimensional occurs at n = 7 with pentagonal bipyramid as the ground state. It was also suggested that Ag_n clusters favor flat cage structures in the size range of n = 8−16, followed by transitions from "empty cage" to "endohedral cage with one central Ag atom" at n = 18, and to "endohedral cage with two central Ag atoms" at n = 22.[14−17] According to theoretical predictions using empirical potentials, nonsymmetric packings and derivatives of the icosahedron structure are energetically favorable for most larger clusters, with a few exceptions.[18−20] For instance, Ag_{38} was predicted to be a truncated octahedron with cubic close packing. Ag_{55} and Ag_{147} are perfect icosahedra in I_h symmetry with smaller Ag_{13} and Ag_{55} cores, respectively.[21−23]

To rationalize the structure and stability of silver clusters, various theoretical models have been developed, such as the electronic shell model based on the spherical jellium approximation, which reveals the "magic numbers" of n = 2, 8, 18, 20, 34, etc.,[24−27] and the classic Wulff construction principle for icosahedral clusters with n = 13, 55, 147, etc.[28] Generally speaking, these theoretical models offer a reasonable explanation of the size evolution behavior in experimental photoionization mass spectra, ionization potentials, electron affinities, and relative kinetic rates for chemical reactions.[29−31] However, there is one exception to the well-known "magic sizes", i.e., a sharp decrease in the vertical ionization potential from Ag_{48} to Ag_{49} in photoionization measurements of neutral clusters.[31] Accordingly, distinct inert-to-active transitions from Ag_{47}^- to Ag_{48}^- in the relative kinetic rate for adsorption of O_2 on Ag_n^- clusters and from Ag_{49}^+ to Ag_{50}^+ in the O_2 adsorption on Ag_n^+ clusters have been reported by Xing's group[30] and Schmidt et al.,[29] respectively. All these experimental results for neutral, anionic, and cationic clusters indicate that silver clusters with 48 5s valence electrons exhibit enhanced stability corresponding to "magic number" behavior. Another exception

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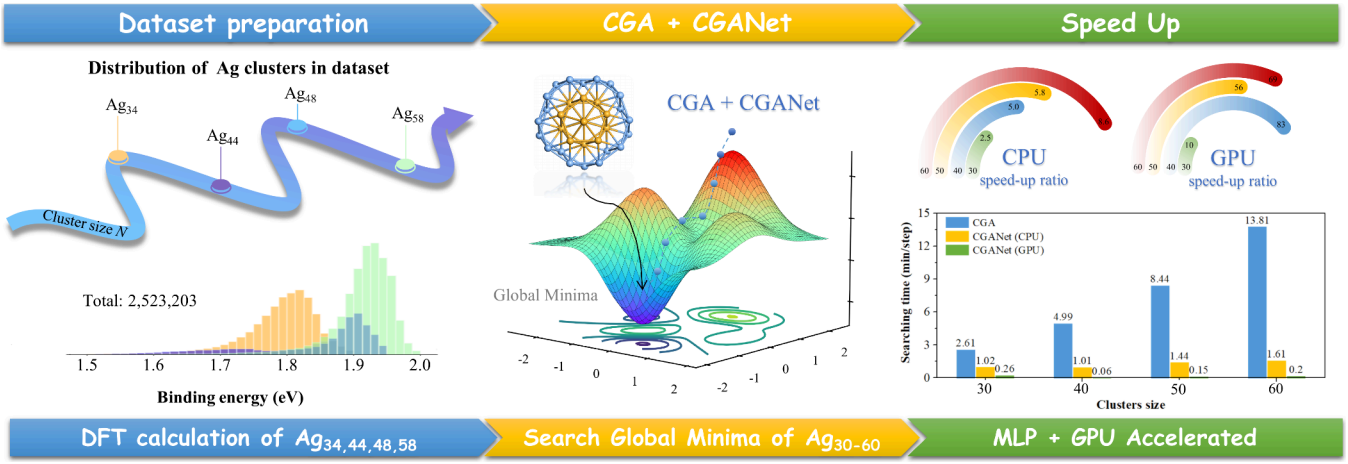


Figure 1. Schematic workflow showing the CGA−CGANet strategy for global search of large-sized silver clusters. (CPU: Intel Xeon Gold 6226R with 36 cores. GPU: NVIDIA GeForce 3090.)

is that in accordance with the spin accommodation mechanism, Ag_{49}^- is expected to exhibit low adsorption reactivity for O_2. On the contrary, the kinetic rate of Ag_{49}^- observed in Xing's work is highest among all Ag_n^- clusters with $n > 27$, and it is the only size with higher reactivity than that of its two even-sized neighbors. To date, however, the origin of such exceptional stability at Ag_{48} and its neighboring sizes remains a puzzle since it is neither a conventional "magic number" according to the jellium model for valence electrons nor a special size that obeys the classic Wulff construction principle from the geometric point of view.

In order to unveil the stability mechanism of silver clusters with 48 5s valence electrons and abnormal reaction rate of Ag_{49}^-, the first step is to determine the global minimum (GM) structures of Ag_n clusters with relatively large size (e.g., $30 < n < 60$), which is rather computationally expensive at the full level of density function theory (DFT). Instead, the embedded atom method (EAM) has been used for exploring the GM structures of Ag_n clusters up to $n = 100$,[19] but the accuracy of such an empirical potential is questionable. The emerging machine learning (ML) potentials offer an alternative approach to describe the potential energy surface (PES) by fitting a large structure−energy relationship dataset.[32] In this manner, ML models based on convolutional neural network,[33] transfer learning,[34,35] and active learning[36,37] have been successfully applied to speed up the global optimization of clusters such as Ta_n (n = 9−13), Au_n (n = 13), Pt_n (n = 8−14), $(BN)_n$ fullerenes (n = 23−32),[38] and exohedrally functionalized carbon fullerenes.[33] These ML potentials generally show better performance than conventional empirical potentials and have become an increasingly recognized tool for the computer simulation of clusters and nanomaterials. However, these previous studies mainly focused on structural prediction for single-sized or small-sized clusters, probably due to the limitations of size transferability and accuracy. Evidently, a comprehensive ML model taking into account the geometry feature and local atomic environment is desirable for a systematic study of Ag_n clusters in a wide size range in order to disclose the size-dependent evolution of stability and explain the magic number behavior around Ag_{48}.

Inspired by the natural suitability of graph architecture in describing the geometric characteristics of a cluster or molecule, we recently developed a cluster graph attention network (CGANet) to describe the PES of medium-sized Ag_n clusters (n = 14−26).[38] Combined with a homemade comprehensive genetic algorithm (CGA) program[39] for global optimization, a series of unprecedented lowest-energy structures were found for Ag_n clusters with n = 22−26. In this work, we extended CGANet to large-sized Ag_n clusters by acceleration with a graphics processing unit (GPU) and symmetry constraint. CGANet can accurately predict the binding energy and force of silver clusters with a root-mean-square error of 2.13 meV/atom and a mean absolute error of 29.1 meV/Å, respectively. Benefiting from the improved efficiency and accuracy, the GM structures of Ag_n clusters with n = 30−60 have been determined, revealing a multilayer icosahedral-based growth pattern. In particular, the exceptional stability at the Ag_{48} cluster has been well-interpreted in terms of both geometric and electronic factors.

Inspired by the theory of evolution, the genetic algorithm (GA) is a well-known heuristic method to mimic the evolution process of "natural selection" in biology. As a global optimization approach, it can avoid being trapped in local minima by a series of mating and mutation operations. Previously, our group developed a GA-based program, namely, CGA,[39] which can be incorporated with DFT calculations and allows efficient yet accurate searches for the PESs of various atomic and molecular clusters. Therefore, the combined CGA−DFT strategy is suitable to construct the dataset of selected sized clusters by sampling a broad region of the PES.

Very recently, we developed a graph network for atomic clusters, namely, CGANet,[38] which is built by aggregation layers of vertex, edge, bond angle, and force information using the attention mechanism. CGANet only requires the type of element and coordinates of every atom as input and can predict energy, force, and electronic properties with satisfactory accuracy comparable to DFT calculations. As a proof of concept, we performed global optimization of medium-sized Ag_n clusters (n = 14−26) using a combination of CGA and CGANet, i.e., the CGA−CGANet strategy. The GM structures for n = 14−21 reported in the literature were reproduced, while entirely new lowest-energy structures were obtained for n = 22−26. Motivated by this success, in the present work we employed the same architecture of the CGANet model and retrained the networks for larger Ag_n clusters (n = 30−60). The workflow of the CGA−CGANet

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strategy for a global search of large-sized silver clusters is shown in Figure 1. To further speed up the calculations, we implemented GPU acceleration and symmetry constraints in CGANet. The Broyden−Fletcher−Goldfarb−Shanno (BFGS) algorithm,[40] which optimizes the cluster structure by calling for the energy and force functions provided by the well-trained CGANet, has been rewritten to run on both CPU and GPU. With GPU acceleration, the speedup rate for CGA−CGANet compared to the CGA−DFT method reaches up to 2 orders of magnitude (Figure 1). During geometry optimization, it is also optional to constrain the symmetry and fix the coordinates of selected atoms. When constrained symmetry is applied, the atomic coordinates are adjusted after each iteration to maintain the presumed symmetry.

For structural prediction of clusters, training a good ML model relies on a good dataset, which should contain adequate samples either close to or far away from the local minima on the PES. To cover the geometric complexity and diversity of large-sized Ag_n clusters, we performed CGA−DFT global search on four selected sizes (Ag_{34}, Ag_{44}, Ag_{48}, and Ag_{58}) with symmetry constraints of C_1, C_2, C_3, C_i, and C_s point groups, i.e., a total of 20 independent searches. For each cluster and constrained point group symmetry, the CGA search lasted for 5000 iterations, and all child clusters were fully optimized with DFT as implemented in the DMol^3 program[41] using a double numerical basis including d polarization function (DND), the Perdew−Burke−Ernzerhof (PBE) functional for the exchange−correlation interaction,[42] and the DFT semicore pseudopotential to treat the core. In this work, the binding energy of a Ag cluster is defined by $E_b = E_{atom} - E_{cluster}/N$, where E_{atom} and $E_{cluster}$ are the total energies of an individual Ag atom and the Ag cluster, respectively, and N is the number of atoms in the cluster. By definition, a positive binding energy means the formation of a cluster from individual atoms is exothermic. By collecting all the intermediate structures with positive E_b during geometry optimization, the total size of the dataset reached more than 2.5 million, that is, 813573, 122302, 453471, and 1133857 data for Ag_{34}, Ag_{44}, Ag_{48}, and Ag_{58}, respectively.

As mentioned above, only four out of 31 sizes between Ag_{30} and Ag_{60} were selected to prepare the dataset, which were then used to train the energy and force networks of CGANet. As a matter of fact, the binding energy of the atomic cluster usually increases with size and gradually approaches the bulk limit, as signified by the average binding energies of Ag_{34}, Ag_{44}, Ag_{48}, and Ag_{58} in Table 1. Consequently, our test calculations indicated that mixed-size training yields less accurate networks than those trained independently for the dataset of each individual size. In practice, we trained the CGANet on selected single-sized datasets and then applied the well-trained model to search for the GM structures of Ag_n clusters of neighboring sizes. During the training process, a batch size of 64 and a learning rate of 0.001 were utilized.

Table 1. Mean Values of Binding Energy ($\langle E_b \rangle$) for Datasets with Different Sizes, RMSEs for Energy ($RMSE_E$), and MAEs for Force (MAE_F)

	Ag_{34}	Ag_{44}	Ag_{48}	Ag_{58}
$\langle E_b \rangle$ (eV)	1.78	1.84	1.84	1.91
$RMSE_E$ (meV/atom)	3.71	3.91	3.36	2.13
MAE_F (meV/Å)	37.5	40.7	29.1	29.8

As a representative, the test errors of energy and force for Ag_{34} converged rapidly to 3.71 meV/atom and 37.5 meV/Å after 100 and 50 epochs, respectively (see Figure 2). The convergence behavior of test errors for Ag_{44}, Ag_{48}, and Ag_{58} clusters are plotted in Figure S1 and summarized in Table 1. For all systems, the root-mean-square error (RMSE) of the energy is less than 4 meV/atom. In other words, CGANet effectively establishes a direct mapping from atomic structure to energy/force that is almost as accurate as DFT calculations. Also note that the current test errors are lower than previous values for medium-sized Ag_n (n = 14−26) clusters (RMSE of 5.4 meV/atom for binding energy and MAE of 42.3 meV/Å for force),[38] Au_n (n = 13) clusters (MAE within 30 meV/atom for energies and 40 meV/Å for force), and Pt_n (n = 8−14) clusters (RMSE within 100 meV for energies) using deep neural network.[35,37]

To validate the transferability and reliability of the CGA−CGANet strategy across a wide range of cluster sizes, we carried out the following three testing tasks. First, taking Ag_{34} as an example, we combined the well-trained CGANet-Ag_{34} energy and force models with the BFGS algorithm to optimize the candidate structures from global search using the same deep learning models. As shown in Figure 2c, the data points are mainly distributed around the diagonal, meaning that the energy differences predicted by DFT and CGANet for the final population structures have a MAE loss of 3.49 meV/atom, which is much smaller than a previous value of 20 meV/atom for Li_n (n = 3−10) clusters also based on graph theory.[43] As the second test, we searched the highly symmetric Ag_{38} and Ag_{55} clusters using the well-trained models of CGANet-Ag_{34} and CGANet-Ag_{58}, respectively. We successfully obtained the well-known structures of Ag_{38} and Ag_{55} with O_h and I_h symmetry, respectively.[21,22] Third, we performed a CGA−CGANet search on intermediate sizes of n = 45, 46, and 47, which are covered by both the CGANet-Ag_{44} and CGANet-Ag_{48} models. After independent global searches, it turned out that the two models trained with different datasets yielded identical GM structures for Ag_{45}, Ag_{46}, and Ag_{47} in Figure 3. Moreover, their lowest frequencies do not include any imaginary frequencies (Table S1), providing further evidence of the rationality of these GM structures. All these results demonstrate that our CGA−CGANet strategy is efficient, accurate, and transferable within a certain size range from the trained size (at least ±8 atoms).

We then carried out GM search for silver clusters containing 30 to 60 atoms using four CGANet models, i.e., CGANet-Ag_{34} for n = 30−38, CGANet-Ag_{44} for n = 39−47, CGANet-Ag_{48} for n = 45−52, and CGANet-Ag_{58} for n = 53−60, as illustrated in Figure 3. At each cluster size, we applied symmetry constraints of C_1, C_2, C_3, C_i, and C_s point groups, and each of the independent CGA searches was performed with a population of 192 individuals and 10000 iterations. Afterward, all structures in the final population from CGA−CGANet searches (about 960 isomers for each size) were reoptimized by DFT calculations with higher precision to ensure that the lowest-energy geometries had been reached. It should be noted that the relative energies between the GM and isomers are found to be basis-set-dependent (see Table S2 for the results for Ag_{48} as a representative). However, their energy orderings calculated with double-ζ and larger basis sets are consistent. For each considered size, the relative binding energies obtained by CGANet and DFT calculations are plotted in Figure S2. One can see that the relative order of isomers predicted by

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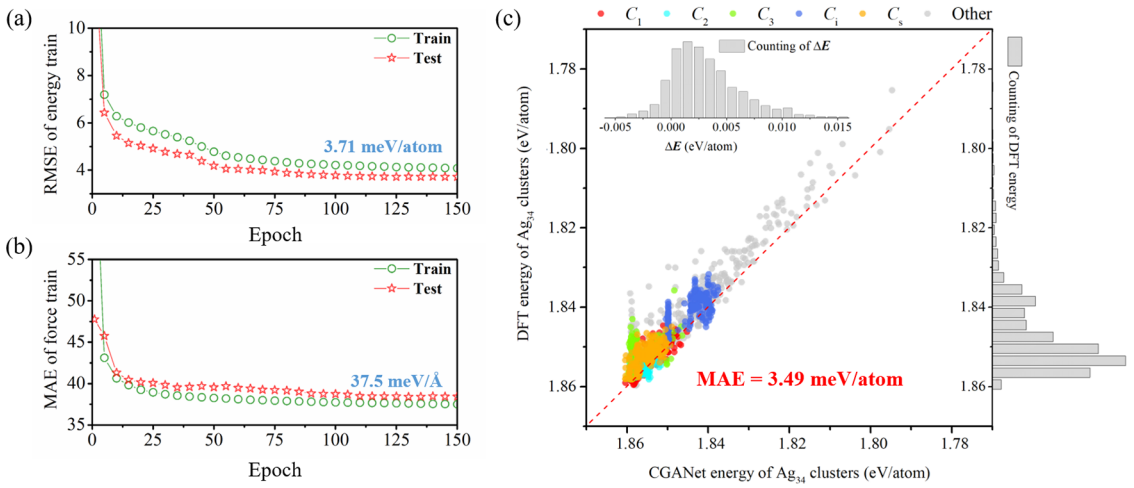


Figure 2. (a) Root-mean-square error (RMSE) of energy and (b) mean absolute error (MAE) of force in Ag_{34} isomers. (c) Binding energies of Ag_{34} isomers predicted by CGANet and DFT calculations, where the gray points are 500 structures randomly selected during the search process and the points with other colors are the isomeric structures in the population after global search within the symmetry constraints of different point groups. The insets illustrate the distribution of the DFT energy and energy difference (ΔE) between the DFT and CGANet.

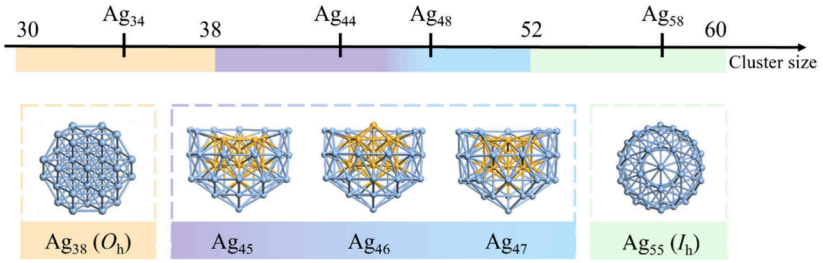


Figure 3. Size segments of CGA−CGANet search strategy and selected sizes (n = 45, 46, 47) for testing the coincidence of different CGANet models.

CGANet generally agrees with that by DFT. Specifically, the GM configurations of eight Ag_n clusters out of 31 sizes are precisely located by CGANet, while the other five DFT ground states fall in the top five lowest-energy isomers by CGANet. All these results demonstrate that the CGA−CGANet strategy allows us to search the PESs of large-sized Ag clusters efficiently with acceptable computational cost. While we cannot ensure the GM structures for all sizes, the documented GM structures can serve as appropriate seeds for further analysis using more accurate methodologies.

The GM structures of Ag_n (n = 30−60) clusters are depicted in Figure 4, and some newly found structural isomers are presented in Figure S3. One can see that the structures of Ag_n clusters exhibit rather complicated growth behavior. Generally, as cluster size increases, the most stable geometry becomes more symmetric, and the interior core (highlighted by yellow) evolves from a Ag_3 triangle to a Ag_{13} icosahedron. In the size range of n = 30−36, oblate core−shell structures with relatively low symmetry (C_1, C_2, C_s) are obtained, and the core architectures include Ag_3 triangle, Ag_4 tetrahedron, Ag_5 trigonal bipyramid, and Ag_5 square pyramid. At a critical size of n = 37, a structural transformation occurs, and the favorable GM structures contain an octahedral Ag_6 core. A highly symmetric truncated octahedron (O_h) is found for Ag_{38}, while the GM structures at n = 37 and 39 can be regarded as its derivatives. As shown in Figure S4, it is interesting to find metastable isomers of Ag_{38} and Ag_{39} with C_{5v} symmetry lying about 0.08 eV higher in energy. Experimentally, Xing et al. confirmed the presence of local order with fivefold symmetry

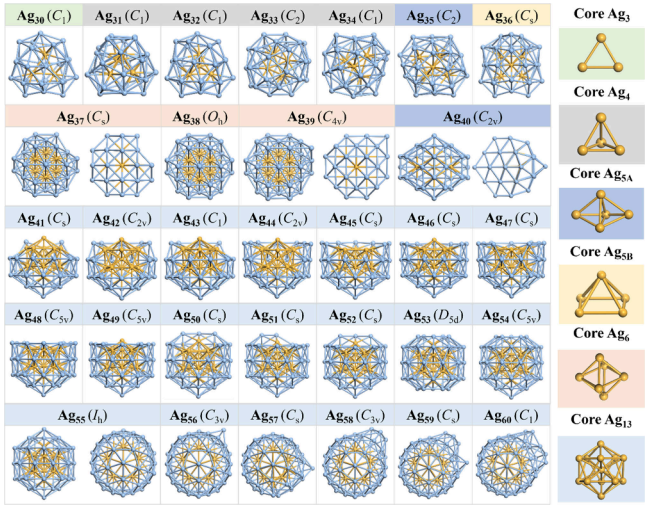


Figure 4. Global minima structures of Ag_n clusters from size 30 to 60. For the left panel, the background color for the title of each Ag_n cluster corresponds to the interior core structure shown in the right panel.

for Ag_n^+ (n = 36−39) by comparison of trapped ion electron diffraction measurements with theoretical structures.[44] As a continuation of these C_{5v} structures, a C_{2v} structure was identified as the GM geometry for Ag_{40}. This structure is formed of two Ino decahedral Ag_{13} cores (C_{5v})[45] packed together by sharing a Ag_7 pentagonal bipyramid and

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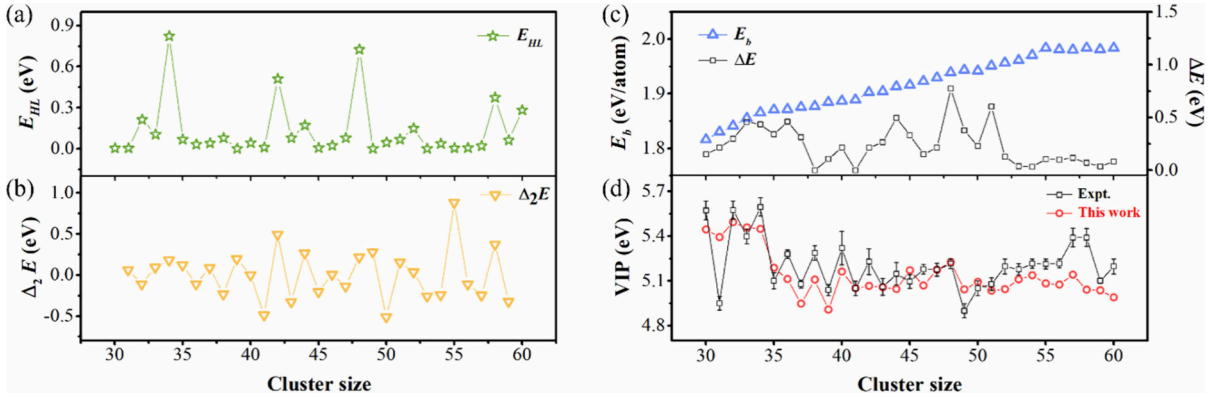


Figure 5. (a) Energy gaps between HOMO and LUMO (E_{HL}), (b) second differences of the energy ($\Delta_2 E$), (c) binding energies (E_b) of GM structures and energy differences (ΔE) calculated at the same PBE/DND level of theory between GM clusters reported in this work and previous work,[19] and (d) vertical ionization potentials (VIPs) for Ag_n (n = 30−60) clusters in GM structures predicted in this work compared with experimental values reported in ref 31.

surrounded by other silver atoms on one side, forming a wedge-shaped cluster that is thicker in the center and thinner on the sides.

For larger Ag_n clusters with n = 41−60, the icosahedron-based motif becomes dominant. As can be seen from Figure 4, all those GM structures (except Ag_{45}, Ag_{47}, and Ag_{48}) adopt a complete Ag_{13} icosahedron as the interior core. Based on this icosahedral core, size-dependent structural evolution in the size range of n = 39−55 can be characterized by a vertical layer-by-layer stacking mode toward a complete Mackay icosahedron of Ag_{55}. For example, the metastable structure of Ag_{39B} (C_{5v}) in Figure S3 can be regarded as 1−5−10−10 (covering the icosahedral Ag_{13} core), while the complete Ag_{55} icosahedron (I_h) can be viewed as 1−5−10−10−10−5−1. Those clusters with sizes in between (i.e., n = 41−55) show an evident stacking pattern that occupies the atomic layers in the outer shell one by one, with only a few exceptions at n = 48 and 53. The newly found GM structure of Ag_{48} has C_{5v} symmetry and five outer-shell layers (1−5−10−10−10); however, its interior core is a truncated Ag_{12} icosahedron instead of Ag_{13}. Adding the missing atom back to the Ag_{13} core leads to the GM structure of Ag_{49}. Similarly, the Ag_{53} cluster exhibits D_{5d} symmetry with two atoms missing in the topmost and bottommost layers of the complete icosahedral Ag_{55}, corresponding to a 5−10−10−10−5 architecture. Beyond Ag_{55}, the icosahedron-based growth pattern continues by capping an adatom, a dimer, a trimer, a trimer and a separate adatom, and a trimer plus a nearby dimer on the Ag_{55} icosahedron, resulting in the GM structures of Ag_{56} (C_{3v}), Ag_{57} (C_s), Ag_{58} (C_{3v}), Ag_{59} (C_s), and Ag_{60} (C_1), respectively.

Benefiting from the well-trained CGANet models, the present GM structures and some metastable isomers are energetically more favorable than those predicted from the EAM empirical potential[19] by 0.1 to 0.8 eV at the same PBE/DND level of theory, as shown in Figures 5, S3, and S4. Moreover, we compared the vertical ionization potentials (VIPs) of Ag_n clusters in GM and metastable structures from DFT calculations with the experimental values in Figures 5 and S4, respectively. Our calculations successfully reproduced the even−odd oscillation of VIPs in the size range of n = 30−43 with a few exceptions, while such an oscillation trend gradually diminishes with increasing number of Ag atoms. Overall speaking, our theoretical data coincide well with experimental ones, with an average deviation of 0.075 eV only.

To characterize the relative stability of a cluster with respect to its neighboring sizes, the second difference of energy, $\Delta_2 E = E_{n+1} + E_{n-1} - 2E_n$, as a function of size n for Ag_n clusters is plotted in Figure 5b. Roughly generally, a peak in $\Delta_2 E$ and the large energy difference between HOMO and LUMO energy (E_{HL}) correspond to high stability of the given sized cluster. In the current situation, we see noticeable peaks at n = 34, 42, 48, and 58 for E_{HL} and at n = 34, 42, 48, 55, 58 for $\Delta_2 E$. Moreover, sharp decreases in VIP value from Ag_{34} to Ag_{35} and from Ag_{48} to Ag_{49} have been observed in both experiment and our calculation (Figure 5d). The high stability of Ag_{34}, Ag_{55}, and Ag_{58} can be easily interpreted by the electronic shell model and geometric factor (i.e., "magic numbers" at n = 34 and 58 and complete Mackay icosahedron at n = 55, respectively), while the exceptional stability of Ag_{48} will be discussed in the following.

As shown in Figure 4, the GM structure of the Ag_{48} cluster exhibits C_{5v} symmetry with a Ag_{12} core covered by 1−5−10−10−10 layers and no imaginary vibrational frequency (Table S1). This structure with C_{5v} symmetry is 0.8 eV more stable than the structure of C_s symmetry with a complete Ag_{13} core and 1−5−10−10−6−3 outer layers reported in a previous study.[19] Furthermore, the photoelectron spectrum of the GM of Ag_{48} anion was simulated and is compared with the experimental spectrum[46] in Figure S5. The first few peaks were well-reproduced by the theoretical calculations (except that the second peak in the experiment is contributed by the coexisting isomer Ag_{48B}), and the vertical detachment energy from the theoretical calculation is very close to the experimental value, with deviation of only 0.01 eV. The excellent agreements clearly indicate that Ag_{48} with C_{5v} symmetry is the true GM configuration and that our CGA−CGANet searching strategy can identify the GM to a satisfactory extent. According to the Clemenger−Nilsson model,[47] the axial distortion of a metal cluster compared to a perfect spherical cage causes a splitting of superatomic orbitals with high angular momentum. As a result, the 1G superatomic orbital splits into five sublevels (Figure 6), all of which are doubly degenerate, except for the $1G_{z^4}$ sublevel, which is singly degenerate at higher energy. Excitingly, the valence electrons in the Ag_{48} cluster fill the low energy levels (four lowest 1G sublevels and the highest 1F sublevels between them). The superatomic orbital arrangement is $1S^21P^61D^82S^21D^21F^{12}2P^61G^41F^21G^4$, which corresponds to the closed electronic subshell with 48 electrons.

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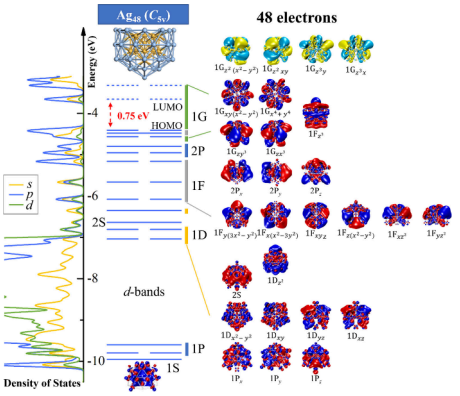


Figure 6. Energy levels and superatomic orbitals for the Ag_{48} cluster. The number of horizontal lines corresponds to the degeneracy of the energy level. The unoccupied orbitals and their energy levels are indicated in different colors and dashed lines.

Note that a large gap up to 0.75 eV is found between HOMO and LUMO due to the splitting of the 1G superatomic orbital. Therefore, even though the number of atoms in the Ag_{48} cluster does not satisfy the completion of the Mackay icosahedron ($N = 1 + \sum_{i=1}^{n}(10i^2 + 2)$),[9] the closed electronic subshell under the Clemenger−Nilsson model still induces remarkable stability. A similar result was found in the gas-phase reaction of Cu_n^+ with O_2, where the oblate Cu_{49}^+ cluster with closed electronic shell exhibited inertness for O_2 adsorption.[48] Also note that although Ag_{49}^- has the same C_{5v} symmetry as Ag_{48} and appears to satisfy the closed electronic shell with 50 5s electrons (see Figure S6), the energy splitting between the 1G sublevels is only 0.067 eV, indicating less stability than Ag_{48}. This provides further insight into the abnormal activity observed in experiments.

In this work, we revisited large Ag_n clusters with $n = 30-60$ using a genetic algorithm code combined with a graph attention network, i.e., CGANet. We identified unprecedented GM geometries for Ag_n clusters, which adopt an icosahedron-based layer-by-layer growth pattern starting from Ag_{40}. The size-dependent binding energy, HOMO−LUMO gap, second energy difference, and ionization potential of these Ag_n clusters are discussed and compared with available experimental data. In particular, the long-lasting puzzle about the exceptional stability of the Ag_{48} cluster from previous experiments has been solved from its C_{5v} geometry and arrangement of superatomic orbitals. It is also noteworthy that the GM search is accelerated by the current deep learning model (i.e., CGANet) by about 2 orders of magnitude with regard to the conventional genetic algorithm strategy, but with accuracy comparable to that of DFT calculations. Finally, our CGA−CGANet strategy is not limited to the global minimum search for Ag clusters but should be applicable to a wide range of other large-sized clusters by revealing the size-dependent structural evolution and rationalizing the experimental results.

■ ASSOCIATED CONTENT

Data Availability Statement

The CGANet code that supports the findings of this study are openly available at the Gitee repository: https://gitee.com/saieuler/cganet.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpclett.4c01953.

Newly discovered lowest-energy structures and metastable structures of Ag_n clusters, acceleration of CGANet, Tables S1 and S2, and Figures S1−S9 (PDF)

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Notes

The authors declare no competing financial interest.

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